Activity 9 — Student Marks Matrix Analysis

1. Problem Statement

Develop a Python program to store marks of three students in three subjects using a matrix. Calculate each student's total marks and average marks.

2. Objectives

- To store student marks in a matrix.
- To calculate total and average marks.
- To understand lists and nested `for` loops.
- To improve basic Python programming skills.

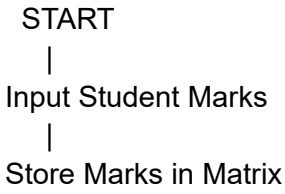
3. Assumptions

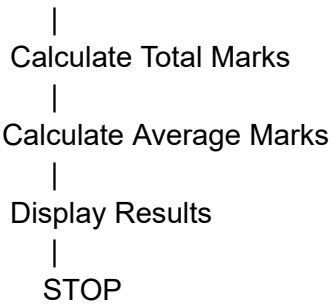
- There are 3 students and 3 subjects.
- Each subject carries 100 marks.
- Marks must be between 0 and 100.
- The program calculates the total and average for each student.

4. Algorithm

1. Start.
2. Create an empty matrix to store marks.
3. Enter marks for three students in three subjects.
4. Check whether each mark is between 0 and 100.
5. Store the marks in the matrix.
6. Calculate the total marks of each student.
7. Calculate the average marks.
8. Display the matrix, total and average.
9. Stop.

5. Flowchart





6. Pseudocode

```
START
CREATE empty matrix

FOR each student
   FOR each subject
      INPUT marks
      IF marks < 0 OR marks > 100
         PRINT "Invalid marks"
         STOP
      ENDIF
      STORE marks in matrix
   ENDFOR
ENDFOR

FOR each student
   total = sum of student's marks
   average = total / 3
   PRINT total and average
ENDFOR

STOP
```

7. Python Source Code

File name: `activity9.py`

```
marks = []

for i in range(3):
   student_marks = []

   print("Enter marks for Student", i + 1)

   for j in range(3):
      mark = float(input("Enter subject marks: "))
```

```
        if mark < 0 or mark > 100:
            print("Invalid marks")
            quit()

        student_marks.append(mark)

    marks.append(student_marks)

print("\nStudent Marks Analysis")

for i in range(3):
    total = sum(marks[i])
    average = total / 3

    print("Student", i + 1)
    print("Marks:", marks[i])
    print("Total:", total)
    print("Average:", round(average, 2))
```

8. Test Cases

Student	Subject 1	Subject 2	Subject 3	Total	Average
1	80	75	90	245	81.67
2	70	85	80	235	78.33
3	90	95	88	273	91.00

Test Input	Expected Result
Marks: 80, 75, 90	Total: 245; Average: 81.67
Marks: 100, 100, 100	Total: 300; Average: 100
Marks: -5	Invalid marks
Marks: 105	Invalid marks

9. Output Screenshot and Conclusion

Sample Output:

```
Enter marks for Student 1
Enter subject marks: 80
Enter subject marks: 75
Enter subject marks: 90
```

Enter marks for Student 2
Enter subject marks: 70
Enter subject marks: 85
Enter subject marks: 80
Enter marks for Student 3
Enter subject marks: 90
Enter subject marks: 95
Enter subject marks: 88

Student Marks Analysis
Student 1
Marks: [80.0, 75.0, 90.0]
Total: 245.0
Average: 81.67
Student 2
Marks: [70.0, 85.0, 80.0]
Total: 235.0
Average: 78.33
Student 3
Marks: [90.0, 95.0, 88.0]
Total: 273.0
Average: 91.0

Output Screenshot: Run `activity9.py` in Python IDLE or VS Code. Capture the inputs and output, and save the screenshot as `activity9_output.png`.

Conclusion: This program uses a matrix, nested loops and arithmetic operations to calculate student totals and averages.

INPUT:

```
marks = []

for i in range(3):
    student_marks = []

    print("Enter marks for Student", i + 1)

    for j in range(3):
        mark = float(input("Enter subject marks: "))

        if mark < 0 or mark > 100:
            print("Invalid marks")
            quit()

        student_marks.append(mark)

    marks.append(student_marks)

print("\nStudent Marks Analysis")

for i in range(3):
    total = sum(marks[i])
    average = total / 3

    print("Student", i + 1)
    print("Marks:", marks[i])
    print("Total:", total)
    print("Average:", round(average, 2))
```

OUTPUT:

```
Enter marks for Student 1
Enter subject marks: 80
Enter subject marks: 75
Enter subject marks: 90
Enter marks for Student 2
Enter subject marks: 70
Enter subject marks: 85
Enter subject marks: 80
Enter marks for Student 3
Enter subject marks: 90
Enter subject marks: 95
Enter subject marks: 88

Student Marks Analysis
Student 1
Marks: [80.0, 75.0, 90.0]
Total: 245.0
Average: 81.67
Student 2
Marks: [70.0, 85.0, 80.0]
Total: 235.0
Average: 78.33
Student 3
Marks: [90.0, 95.0, 88.0]
Total: 273.0
Average: 91.0
```